

Directing Depolymerization of PET with Subcritical and Supercritical Ethanol to Different Monomers through Changes in Operation Conditions

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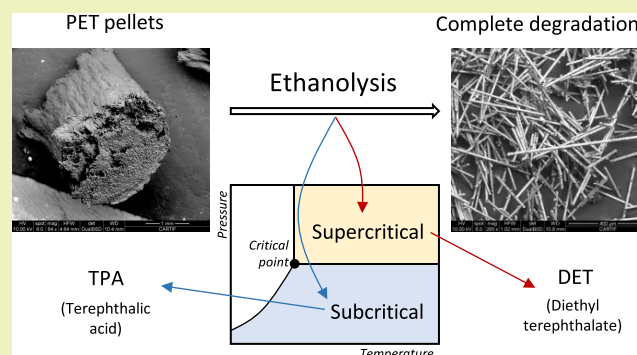
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Supporting Information

ABSTRACT: Recycling of poly(ethylene terephthalate) (PET) wastes has become an urgent need since huge amounts of this plastic are accumulated in the environment. Unlike other chemical recycling methods, depolymerization of PET with supercritical ethanol is based on the use of a nontoxic green solvent. However, the viability of ethanolysis further depends on finding mild pressure and temperature conditions that do not interfere with the degradation efficiency. This challenge was faced in this study by performing ethanolysis of PET under unexplored pressure, temperature, reaction time, and weight ratio PET to ethanol (% wt) conditions. The variation of ethanolysis conditions revealed that besides the expected diethyl terephthalate (DET) and ethylene glycol (EG), other degradation products were also obtained, such as terephthalic acid (TPA). The exploration of the pathways involved allowed directing the process toward different degradation monomers by changing ethanolysis conditions. Long supercritical treatments completely degraded PET and essentially led to the formation of DET and EG since the primarily formed TPA reacted with ethanol under these conditions to produce further DET. Subcritical pressures and temperatures, short reaction times, and high % wt did not always result in complete degradation of PET but increased TPA formation, the main monomer regarded as the building block for PET repolymerization.

KEYWORDS: plastic waste, nontoxic solvent, supercritical fluid, recycling, terephthalic acid, diethyl terephthalate



INTRODUCTION

With an estimated demand of about 4 million tons in 2019 only in the European Union and Norway, poly(ethylene terephthalate) (PET) is the most widely used condensation polymer in the world, together with polyurethanes.^{1,2} The extensive use of this plastic relies on its excellent physical and chemical properties that allow using PET in diverse applications, from food and beverage packaging to textile and automotive manufacturing.³ Due to its widespread use and its resistance to atmospheric and biological agents, huge amounts of PET wastes are being accumulated in the environment, posing a threat to ecosystems and human health. This situation has made the recycling of PET wastes a necessity.

PET can be recycled through re-extrusion, mechanical treatments, chemical treatments, or energy recovery. Chemical recycling with solvents is interesting for condensation polymers vulnerable to solvolytic chain cleavage such as PET.⁴ Solvent exposure depolymerizes plastic complex chains into their raw monomers. The obtained monomers can be used either as building blocks to repolymerize new plastic chains or as chemical feedstocks in any other application. Thus, plastic waste seems to be a promising platform for the production of valuable chemicals and fuels.⁵ PET can be depolymerized by

different solvents. Glycolysis, the oldest method in the field, consists of exposing plastic waste to ethylene glycol (EG) at 180–250 °C and pressures up to 6 bar to obtain bis(2-hydroxyethyl) terephthalate (BHET).³ Beyond glycolysis, one of the most common chemical methods to recycle PET is alcoholysis employing methanol to produce dimethyl terephthalate (DMT) and EG. Process conditions investigated range from 20 to 200 bar and 180 to 300 °C. Subcritical methanolysis below its critical constants of 80.4 bar and 239.6 °C usually needs transesterification catalysts to accelerate the process.⁴ Thanks to the larger densities and higher kinetic energies of fluids above their critical points, the rate of depolymerization was found to be faster when methanol was used under supercritical conditions.^{6–10} As a consequence, this supercritical process has attracted much attention as the

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alternative with the greatest potential within chemical recycling of PET.³ Together with the necessity for high pressure and temperature, the greatest drawback of supercritical methanolysis of PET is the toxic character of methanol. This risk can be avoided if ethanol is used instead of methanol. Despite the great potential and safer character of ethanolysis of PET, this process has been rarely reviewed. De Castro et al. first explored depolymerization of PET in supercritical ethanol at 255 °C, initial pressure of 115 or 165 bar, and reaction times between 5.0 and 6.5 h.^{11,12} They found that PET was completely depolymerized and the main products obtained were diethyl terephthalate (DET), EG, and BHET. Later efforts by the same research group have been focused on the search for catalysts such as ionic liquids^{13,14} or metal oxides¹⁵ that lead to faster degradation kinetics under the same temperature and pressure conditions.

The current work aims at expanding the available knowledge about this safe and efficient depolymerization process of PET by broadening the assayed ranges of temperature, pressure, reaction time, and weight ratio of PET to ethanol (% wt) in the absence of any catalyst. The starting objective was to explore the effect of all of these parameters on degradation kinetics and the amounts of produced monomers, an issue that has received scarce or no attention so far. The screening of wide ranges of conditions allowed investigating degradation of PET with subcritical ethanol ($P_c = 60.4$ bar, $T_c = 240.9$ °C) for the first time. From an upscaling point of view, a decrease in temperature and above all pressure without suffering remarkable efficiency losses is critical. Exploring softer conditions consequently seemed mandatory regarding the technical feasibility of this method. Testing unexplored conditions revealed new information about how ethanolysis of PET proceeds, which products are obtained when ethanol is used under different physical states, and a better understanding of the kinetics and mechanisms involved. This contribution may lead to a new circular process consisting of reducing PET waste disposal in the environment while upgrading it as a feedstock to obtain monomers, which could be used to synthesize new PET or other high-value-added chemicals.

■ EXPERIMENTAL SECTION

Commercial pellets purchased from Merck consisted of 70% pure PET and 30% glass fibers as reinforcers. Ethanol of HPLC quality was purchased from Scharlau. Analytical-standard-grade DMT, terephthalic acid (TPA), and EG were purchased from Merck, and DET was from KeyOrganics.

Ethanolysis assays were carried out in a 316 stainless steel Novoclave Büchi 300 autoclave by Büchi AG with an operating volume of 340 cm³ placed inside a heating jacket (electrical resistances) with an internal refrigeration circuit. The mass of ethanol charged was calculated through its compressibility factor at the desired working temperature and pressure. Preliminary assays confirmed that the pursued pressures were achieved for the calculated masses of ethanol. The mass of pure PET charged depended on the calculated mass of ethanol and the desired working % wt, as shown in eq 1

$$\% \text{ wt} = \left(\frac{\text{mass of pure PET (g)}}{\text{mass of ethanol (g)}} \right) \times 100 \quad (1)$$

The reactor was subsequently assembled and purged with a N₂ flow to remove the presence of oxygen. After the purge, the autoclave was heated to the desired temperature at a heating rate of about 10 °C min⁻¹. This heating rate was maintained up to the highest assayed temperature; thus, the extra heating time between subsequent

temperatures was typically shorter than 3.0–3.5 min. Pressure was simultaneously measured with an analogical and a digital manometer. Temperature and pressure measurement accuracies were ± 0.1 °C and ± 0.1 bar, respectively. Once the reaction conditions were reached, temperature and pressure variabilities were ± 3.0 °C and ± 1.0 bar, respectively. To stop the reaction, heating was stopped and cold water was circulated through the refrigeration circuit. Cooling from 300 to 200, 200 to 100, and 100 °C to ambient temperature, respectively, took about 10, 20, and 60 min. Reaction products were recovered after cooling. The product obtained in the reaction was a mixture of a solid phase and a liquid phase; no gas phase was formed. Both phases were separated by vacuum filtration.

The separated wet solids were dried at 70 °C until constant mass. Besides being weighed, the collected solids were analyzed by differential scanning calorimetry (DSC) in a Q100-TA instrument equipment (25–500 °C with a heating rate of 10 °C min⁻¹ and a N₂ flow of 50 cm³ min⁻¹) and scanning electron microscopy (SEM) in a Quanta 200-FEI Company. The dry solids were a mixture of nonreacted PET, TPA, and glass fiber. The mass of TPA contained in the solids was quantified by spectrophotometer analysis at 240 nm after extraction with NaOH 1.25 M (Nanodrop 2000 from Thermo Scientific). Quantification was performed according to an external standard. Some commercial pellets exposed to this same NaOH extraction did not suffer from any weight loss, and no peak was observed in the spectrophotometric profiles at 240 nm. This confirmed that PET was not degraded by this NaOH extraction.

Once the TPA in the solid was quantified, the PET degradation efficiency could be calculated, as shown in eq 2

degradation efficiency (%)

$$\begin{aligned} &= \left[\frac{(\text{initial mass of pure PET} - \text{mass of nonreacted pure PET})}{\text{initial mass of pure PET}} \right] \times 100 \\ &= (1 - [(\text{mass of dry collected solid} - \text{mass of TPA in the solid} \\ &\quad - \text{mass of glass fiber in the solid}) / \text{initial mass of PET}]) \times 100 \end{aligned} \quad (2)$$

where mass of the glass fiber in the solid was 30% of the initial mass of commercial pellets.

The separated liquid contained phthalates and glycols soluble in ethanol. Phthalates were identified by HPLC-MS in an HPLC Agilent 1100 equipment coupled to a mass spectrometer Trap XCT with an Agilent C18 column; quantification was made by HPLC-DAD according to the external standard method. Glycols were identified and quantified by gas chromatography–mass spectrometry (GC–MS) Shimadzu GC2010 with a DB-WAX-UI column.

The production yield of each individual species produced in ethanolysis was defined, as shown in eq 3

$$\text{production yield} = \frac{\text{mass of species produced (g)}}{\text{mass of pure PET degraded (g)}} \quad (3)$$

Reproducibility was checked and variability was depicted in graphics through error bars. Further information about the analytical procedures and the calculation of error bars is available in the Supporting Information.

■ RESULTS AND DISCUSSION

Ethanolysis of PET Product Analysis. Initial experiments at 40 bar/255 °C for 90 min provided a first approach to degradation product distribution in the liquid and solid phases. The major species detected in the liquid phase were DET, DMT, and EG. Minor concentrations of other phthalates (BHET, ethyl-methyl-terephthalate, and another terephthalate whose two chains summed 10 carbon atoms) and glycols (butanediol and diethylene glycol) were also detected.

The collected solids were characterized by DSC and compared to pure PET, as shown in Figure 1.

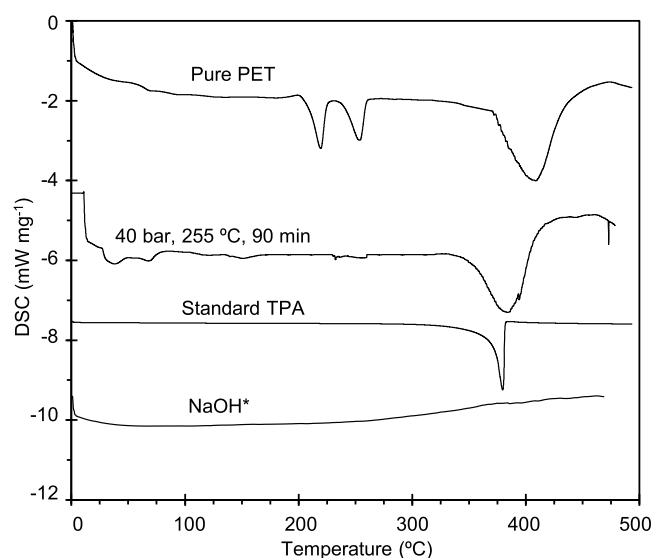


Figure 1. DSC thermograms of pure PET, solids collected after 40 bar/255 °C ethanolysis, standard TPA, and solids from ethanolysis after NaOH extraction (NaOH*).

DSC thermogram of pure PET showed three peaks. The peak at 254 °C was characteristic of PET melting. A second peak appeared at 220 °C. Multiple peaks in calorimetry

analyses of this kind of samples are attributed to the thermal history of the polymer.¹⁶ Namely, peaks with low melting points are usually attributed to the formation of crystals at low temperatures, which leads to an organization in thinner and less thermodynamically stable lamellae structures.¹⁷ The third peak appeared at high temperatures, 410 °C, and was related to PET boiling.

When pure PET was treated at 40 bar/255 °C for 90 min, the low and the mid-temperature peaks disappeared and a high-temperature peak was again observed but at 380 °C. No characteristic peak of PET was observed, so the plastic seemed to have been completely degraded. The new peak might be caused by some degradation product that, once formed, remained retained in the solid. Some samples were extracted with ethanol, and the extracts were analyzed by HPLC-DAD, but no signal was observed, which discarded the presence of DMT and DET in the solid. The DSC thermogram of standard TPA revealed that this peak at 380 °C was probably caused by the presence of this species in the solid. This statement was confirmed by spectrophotometric analyses after NaOH extraction of the postreaction solid since the spectra of standard TPA diluted in NaOH solution and the extract from the sample agreed (see the Supporting Information, Figure S1). The thermogram of the solid from ethanolysis after NaOH extraction did not show any peaks, all TPA were extracted, and the remaining solid was exclusively formed by

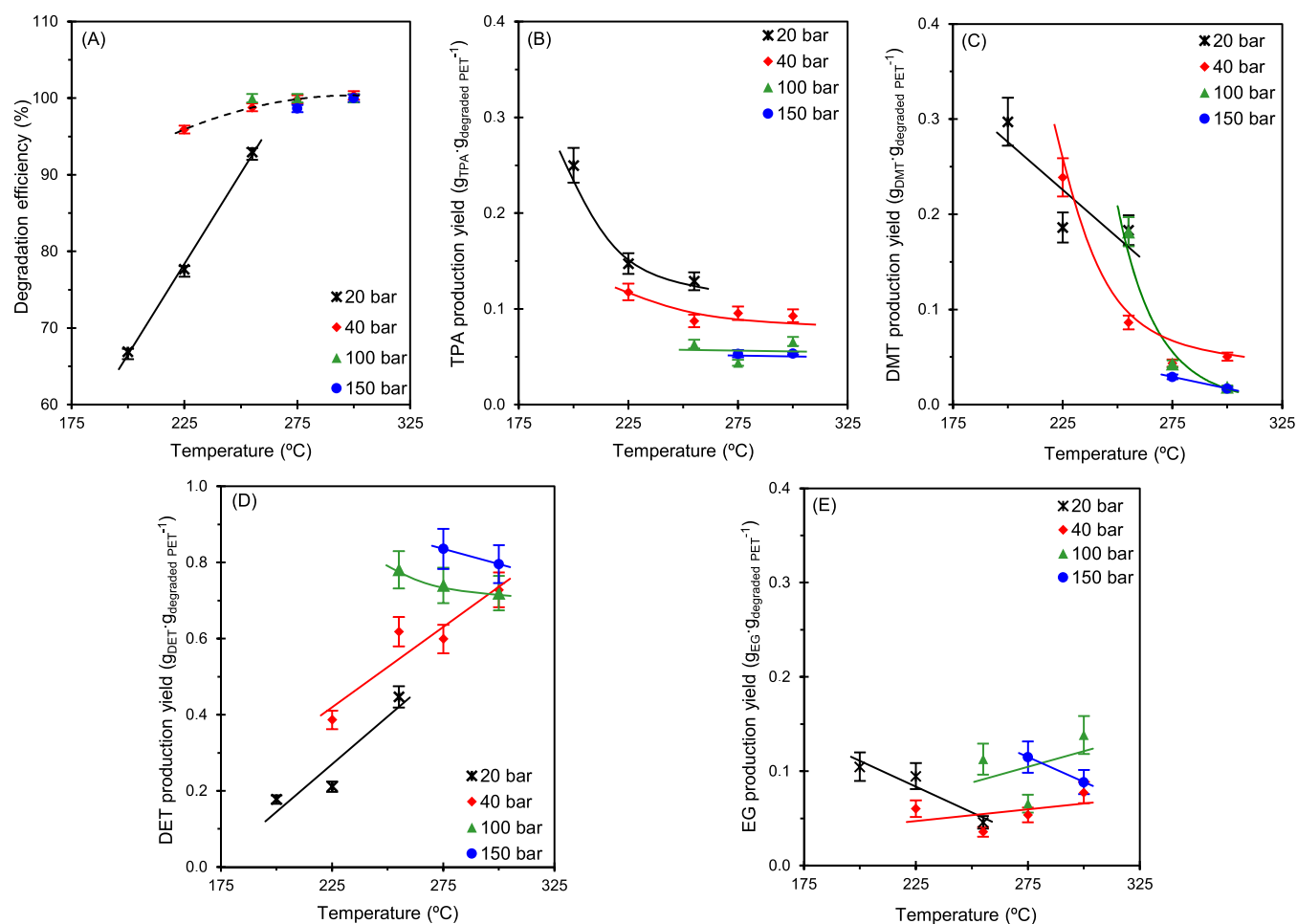
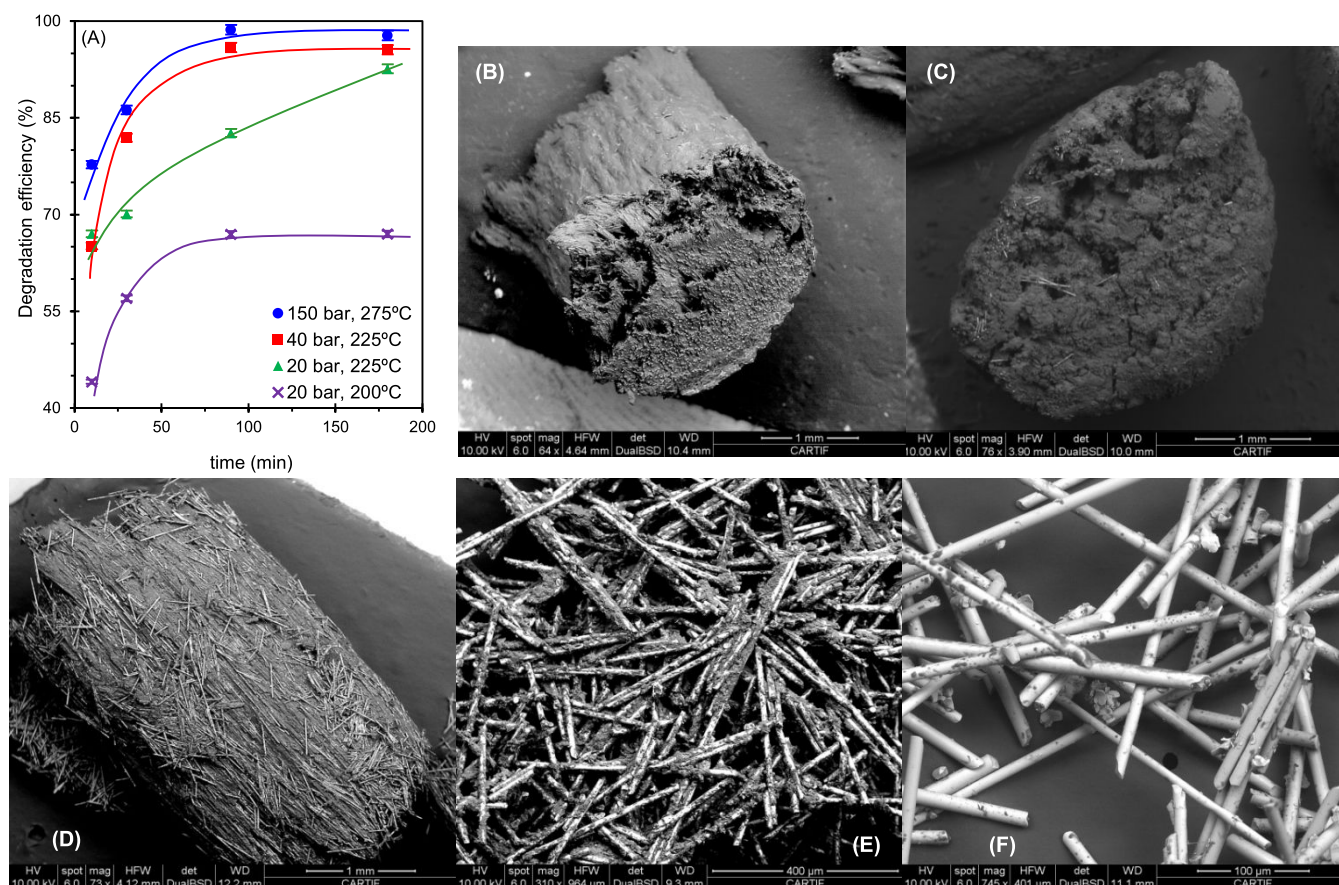


Figure 2. Effect of temperature and pressure on (A) PET degradation efficiency and (B) TPA, (C) DMT, (D) DET, and (E) EG production yields for 90 min ethanolysis treatments (5% wt).

Table 1. Degradation Efficiency and Products Obtained in the Ethanolsysis of PET at 275 °C for 30 min (5% wt)

pressure (bar)	degradation efficiency (%)	TPA (g/100 g _{PET degraded})	DMT (g/100 g _{PET degraded})	DET (g/100 g _{PET degraded})	EG (g/100 g _{PET degraded})	products sum (g · g _{PET degraded} ⁻¹)
40	94.21 ± 0.50	9.40 ± 0.69	12.7 ± 1.08	67.1 ± 4.21	4.73 ± 0.69	0.9393
100	93.45 ± 0.50	7.44 ± 0.54	7.75 ± 0.66	77.6 ± 4.87	5.09 ± 0.74	0.9788
150	90.55 ± 0.48	7.29 ± 0.46	3.92 ± 0.25	82.7 ± 5.80	12.5 ± 1.90	1.0641
200	87.95 ± 0.47	5.74 ± 0.45	2.96 ± 0.33	92.4 ± 5.19	13.4 ± 1.94	1.1450

**Figure 3.** (A) Effect of time on PET degradation efficiency. SEM micrograph images of different samples: (B) commercial pellets, (C) pellets after ethanolsysis at 20 bar/200 °C for 10 min, (D, E) pellets after ethanolsysis at 40 bar/225 °C for 10 min, and (F) pellets after ethanolsysis at 150 bar/275 °C for 90 min.

structural glass fiber, which produced no signal in the calorimetric analyses below 500 °C.

As far as we are concerned, TPA and DMT generation in ethanolsysis of PET has not been reported so far. The four predominant species identified can be used to synthesize new PET, but the repolymerization stage will depend on the monomer distribution after depolymerization. To be able to direct ethanolsysis toward different products, it becomes necessary to investigate qualitatively and quantitatively which of these products are formed depending on depolymerization conditions.

Effect of Pressure and Temperature on Ethanolsysis of PET. Figure 2 shows how the PET degradation efficiency and each product yield changed with temperature from 200 to 300 °C and pressure from 20 to 150 bar.

Increasing the reaction temperature noticeably favored degradation. This effect was especially remarkable in the vicinity of the critical temperature, within the range from 200 to 255 °C, because of the higher kinetic energy acquired by the fluid above that temperature: 66.9% PET was degraded at 20

bar/200 °C, whereas 93.0% PET was degraded at 20 bar/255 °C. Above 255 °C (supercritical conditions in experiments at 275 and 300 °C), PET was completely degraded for 90 min treatments. Phthalates were the main products of ethanolsysis of PET, and their yields changed with temperature. At 200 °C, similar amounts of TPA, DET, and DMT were produced. As the temperature increased, decreasing amounts of TPA and DMT and increasing amounts of DET were produced. The most remarkable changes were observed again in the vicinity of the critical temperature, from 200 to 255 °C. As a result, DET accounted for 90–95% of the total phthalates and 75–80% of the total products under supercritical conditions at 300 °C. EG accounted for less than 10–15% of the total product weight, and its concentration did not follow any clear trend as the temperature changed.

Pressure revealed a certain influence on the process. In the explored subcritical region, the compression of ethanol from 20 to 40 bar increased the degradation efficiency. Once inside the supercritical region, the high temperatures were severe enough to completely degrade PET for those 90 min

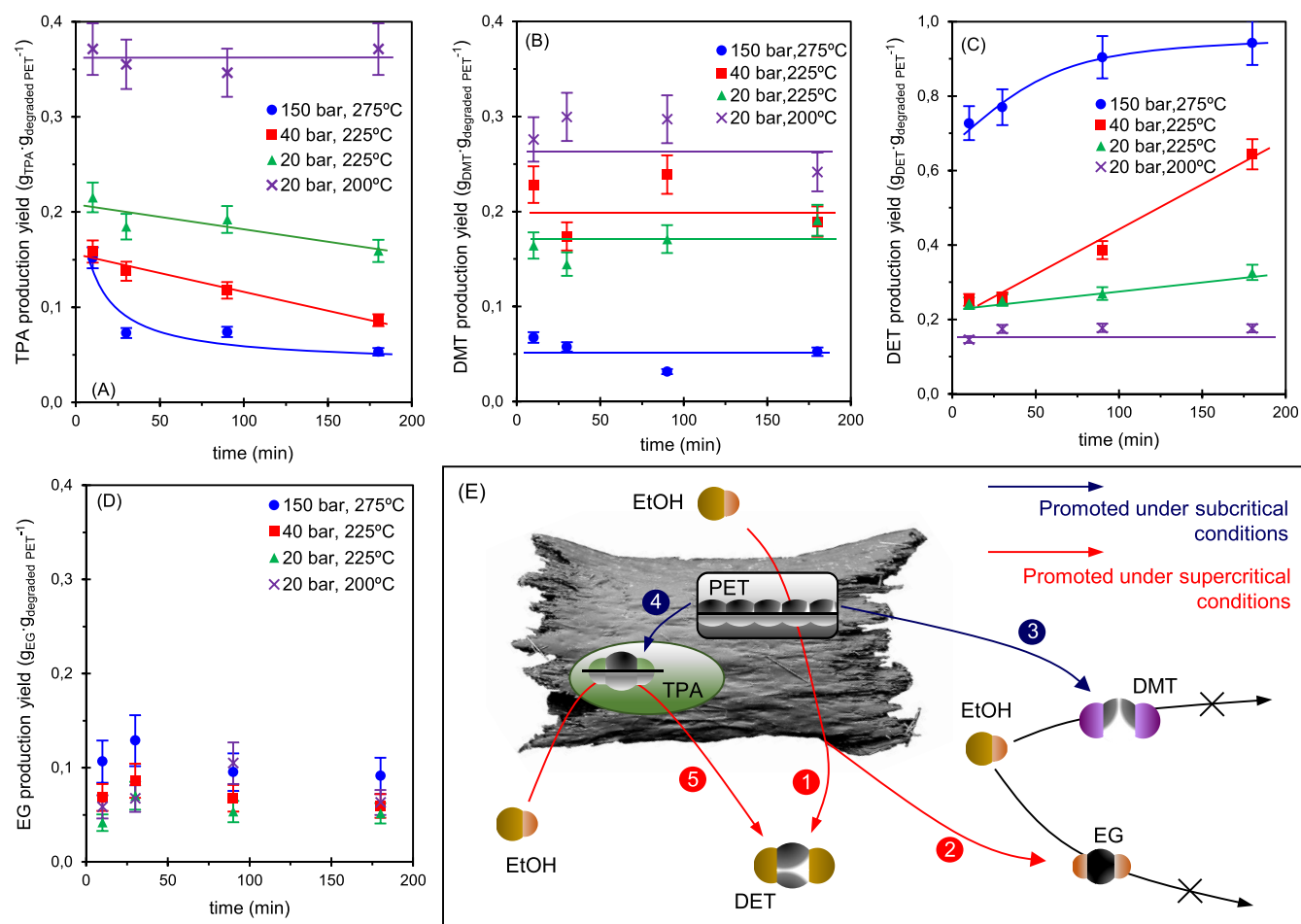


Figure 4. Effect of reaction time on (A) TPA, (B) DMT, (C) DET, and (D) EG production yields for ethanolysis at different temperatures and pressures (5% wt). (E) Potential pathways involved in ethanolysis of PET.

treatments, and no effect of pressure on degradation efficiency was observed. A clear influence of this parameter on the amounts of DMT and EG produced was not observed, but as ethanol was further compressed and passed from the gas to supercritical state, the formation of TPA and DET decreased and increased, respectively. The opposite trends of TPA and DET as temperature and pressure changed provided a first evidence that these two species might be somewhat related. To delve into the influence of pressure on the process, a series of shorter assays (30 min) at 275 °C and pressures from 40 to 200 bar were carried out; see Table 1.

Degradation efficiency decreased slightly when the fluid was progressively compressed from subcritical, 40 bar, to supercritical conditions, 100 bar, and then up to 200 bar. This was probably as a result of the treatment of greater masses of PET as the pressure increased to preserve the same % wt in all of the assays, as discussed in the following sections. DMT and TPA titers diminished, whereas DET production yield sharply increased. This increasing trend was first recorded when the fluid was compressed from the gas state at 40 bar to the supercritical state at 100 bar and was maintained after further compression within the supercritical region to 200 bar.

All of these trends agreed with those in Figure 1. The accurate chosen conditions also clarified that EG production increased with pressure. It must be noted that the sum of the individual production yields increased with pressure and, especially at the highest pressures in the supercritical region,

the sum of the titers of the four main species was well above 1 g of product per gram of PET degraded. At the lowest pressures assayed, this sum was below 1 because all of the minor species were not included in the calculation.

The search for a pressure as low as possible but that leads to an efficient depolymerization is critical since high pressures associated with polymer degradation by ethanolysis are technically challenging. The previous results showed that ethanolysis efficiently depolymerized PET under subcritical conditions, 40 bar/225 °C. This conclusion is relevant since some engineering aspects such as the required alloys and reactor thickness of the reactors become less demanding as the pressure decreases, thereby facilitating the upscaling of pressurized processes.^{4,18}

Degradation Kinetics and Reaction Conditions Influence Ethanolysis of PET. Degradation efficiency was analyzed at four different reaction times under different ethanolysis conditions; see Figure 3. PET decomposition was faster during the first 30 min of reaction, and as ethanolysis was lengthened, the reaction slowed down. The highest degradation efficiency was usually reached after 90 min, and above this reaction time, the profile described a plateau. At 150 bar/275 °C supercritical conditions and 40 bar/225 °C subcritical conditions, essentially all PET was degraded.

Pressure decrease from 40 to 20 bar at 225 °C was detrimental since an almost complete degradation was only achieved after 180 min of reaction, which implied 90 additional

minutes to complete the degradation regarding the 40 bar/225 °C treatment. This delay may be resolved by the use of suitable catalysts.^{13–15} Accelerating subcritical ethanolysis kinetics by catalysis would be another strategy to explore an efficient upscaling of this technique, but the comprehensive use of catalysts entails checking several aspects (efficiency, selectivity, reuse) that are out of the scope of this work. Decreasing the reaction temperature even further below the critical point, 20 bar/200 °C, yielded remarkable efficiency losses: 66.9% PET was degraded after 90 min and further degradation was not observed for longer treatments.

The dependence of PET degradation efficiency on reaction conditions was confirmed by SEM micrograph images; see Figure 3B–F. The complete characterization of each sample by SEM is shown in the Supporting Information, Figures S2–S5.

The two structural components of the commercial pellets are shown in Figure 3B: PET, the main component, presented a resinous appearance, and the tubular shape of glass fibers was also observed. When commercial pellets were exposed to subcritical ethanol at 20 bar/200 °C for 10 min, their shape remained essentially unaltered. Clear evidence of degradation was observed when ethanolysis was carried out at 40 bar/225 °C. Some pellets still preserved their tubular form, but the structural glass fibers were already observed in their outer surface; see Figure 3D. Other pellets had collapsed, and a fraction of the collected solid had a macroscopic dusty appearance. Figure 3E shows that this dust was formed by glass fibers that remained stuck to each other but not strongly enough to preserve the pellet shape. All pellets collapsed in the supercritical ethanolysis at higher temperatures and longer reaction times; see Figure 3D. The collected solid was a dust mainly formed by glass fibers, and almost no solid remained stuck to them. This image confirmed that all PET was degraded under these conditions.

The combined effect of reaction time and ethanolysis conditions on the production yields of the four main depolymerization species was analyzed by offline analytical methods; see Figure 4A–D.

Under 20 bar/200 °C subcritical conditions, about 35% of the degraded PET was transformed into TPA regardless of the reaction time and 25% of the total PET treated was converted to this phthalate after 90 min. TPA formation was interesting because its recovery from the product mixture as a solid (low solubility in ethanol) was relatively easy, and this monomer is commonly used as a raw material for PET repolymerization. TPA yields decreased as temperature and pressure first approached and then surpassed the critical point and the reaction time was lengthened, which agreed with the scarce presence of any solid stuck to the glass fibers collected after 150 bar/275 °C treatments (Figure 3F). Consequently, TPA production was favored by subcritical ethanolysis at low temperature and pressure, under similar conditions to those that lead to TPA formation in another pressurized depolymerization technique as subcritical hydrolysis.^{19,20} The kinetic behavior of TPA was typical of an intermediate product generated as the primary product of ethanolysis but subsequently consumed in secondary reactions. Significant changes along different reaction times were not observed for DMT and EG yields following a primary product behavior. DET production did not change along time during subcritical ethanolysis at 20 bar/200 °C. DET yields at 225 °C treatments for 10 min were higher than at 200 °C, and they increased slightly for longer reaction times. This kinetic trend was clearer

for supercritical 275 °C assays, suggesting that DET was not only a primary degradation product but also probably a product from other secondary reactions occurring in the system under certain conditions.

Based on the quantification of degradation efficiency and the production yields under different ranges of reaction time, temperature, and pressure, potential pathways involved in the ethanolysis of PET and the influence of the physical state of ethanol on them are summarized in Figure 4E. TPA, DMT, DET, and EG were detected from the earliest reaction times under all of the conditions assayed; thus, the four species were primary products. Supercritical ethanolysis boosted DET primary formation (pathway 1), a reaction that also yielded EG (pathway 2).¹³ DMT and TPA primary formations were greater for subcritical conditions (pathways 3 and 4). DMT was dissolved in the bulk media, whereas TPA remained stuck to the solid.

The kinetic behavior in Figure 4 suggested that DMT and EG did not play any role as reagents or products in other secondary reactions and that TPA acted as a reagent in a reaction that probably produced DET. The used bulk media pointed to ethanol as the second reagent in this reaction, pathway 5. The reactivities of standard TPA, DMT, and EG with ethanol at different temperatures, pressures, and reaction times were separately checked to obtain further information about possible secondary reactions (see the Supporting Information, Table S2).

TPA did react with ethanol as suggested in pathway 5: DET was detected as the reaction product, and DMT and EG were not detected in any assay. The conversion of TPA to DET increased with temperature and reaction time, but pressure was the most influential parameter, boosting DET formation: from 16.6% conversion under a 40 bar/275 °C gas state to 56.9% under a 150 bar/275 °C supercritical state. The reactivity of DMT and EG with ethanol was extremely low, agreeing with their primary product behavior.

The compliance of the stoichiometric reaction between PET and ethanol to produce DET and EG was confirmed herein for long supercritical treatments.¹³ Under those conditions, the sum of the individual production yields was above $1 \text{ g}_{\text{products}} \cdot \text{g}_{\text{degraded PET}}^{-1}$ (see Table 1). This meant that supercritical ethanol caused the rupture of PET chains and reacted with the products resulting from breakage. When the ethanolysis conditions were softened to ranges not previously explored, significant amounts of TPA and DMT were detected. Regarding this issue, two reaction paths have been described for methanolysis of PET:^{7–9} pure random degradation, represented by binary scission of bonds at any position along the chain, and the specific reaction of methanol and the chain end that released monomeric species such as DMT. In the system studied herein, simultaneous DET and EG formation from ethanolysis of PET was a chain-end scission, but random scissions leading to DMT and TPA may also happen. The greatest DMT and TPA production yields were recorded under subcritical conditions, and the sum of the individual production yields for those assays was below $1 \text{ g}_{\text{products}} \cdot \text{g}_{\text{degraded PET}}^{-1}$. This meant that subcritical ethanol caused the rupture of PET chains but did not react with the products of that scission (or at least these reactions were much less significant than under supercritical conditions), thus supporting the fact that depolymerization was happening through some phenomena different from chain-end reactions.

All of the previous data allowed foreseeing and understanding which ethanolysis conditions favored each phthalate formation. Short subcritical ethanolysis led to the greatest amounts of TPA and DMT but not to complete degradation of PET. Long supercritical ethanolysis completely degraded PET and mainly produced DET as phthalate due to its formation as a primary and a secondary product. This DET could be recovered from the liquid phase and upgraded to produce new PET by transesterification with EG.²¹ It must be noted that the separation of DET (soluble in ethanol) from the postreaction mixture is more complex than the separation of nonsoluble TPA, and the conditions that maximize DET production were more energetically demanding than subcritical conditions that led to TPA among others. Consequently, supercritical depolymerization cannot be categorically stated as the best option. All of these technical and economic aspects as well as others such as the use of catalysts should be considered to establish the basis of an optimal circular process that produces new pure PET from monomers obtained in ethanolysis of waste PET.

Effect of % wt on Ethanolysis of PET. Checking if ethanolysis was able to depolymerize large amounts of PET under mild conditions was important regarding the viability and efficiency of the method; see Figure 5.

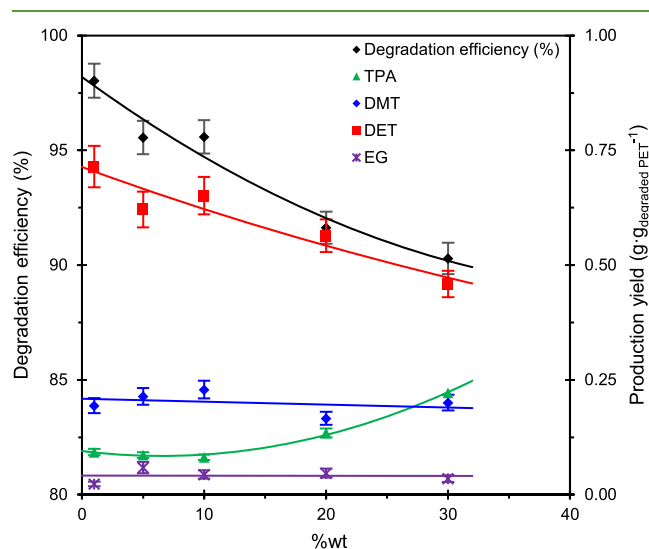


Figure 5. Effect of % wt on PET degradation efficiency and production yields for ethanolysis treatments at 40 bar/225 °C for 180 min.

The decrease in degradation efficiency with % wt was affordable since a 30-fold increase in the mass of PET merely provoked an efficiency loss from 98.0 to 90.3%. Consequently, subcritical ethanolysis could be validated as a highly effective technique to depolymerize PET. These data also explained some results in Table 1. The mass of ethanol charged in the autoclave for supercritical assays at 200 bar was about 6 times greater than for subcritical assays at 40 bar because of the change in the fluid density with pressure. As all experiments in Table 1 were carried out at 5% wt, the mass of PET treated at 200 bar was consequently 6 times greater than at 40 bar. Figure 5 shows that the degradation efficiency decreased as larger masses of PET were treated, thereby explaining why the lowest efficiency in Table 1 was recorded for the 200 bar assay.

Modifying % wt was also revealed as an interesting strategy to direct the process to different depolymerization products. TPA production yields were about $0.10 \text{ g}_{\text{TPA}} \cdot \text{g}_{\text{degraded PET}}^{-1}$ from 1 to 10% wt, started rising from 10 to 20% wt, and doubled for 30% wt assays. As a result, the amount of TPA produced at 30% wt was 65-fold greater than at 1% wt. The net amounts of the other products increased for the highest % wt (even for DET despite the diminishment in its production yield) although the increase in TPA production was the most remarkable. These data confirmed that ethanolysis of PET could be directed toward different monomers by changing the degradation conditions.

CONCLUSIONS

Subcritical and supercritical ethanolysis was proved as an efficient method to depolymerize PET and obtain monomers suitable for its exploitation as a raw material to polymerize new PET or other chemicals. The greatest shortcoming of this method is the necessity for high pressures, but it was concluded that a subcritical 40 bar pressure was enough to efficiently depolymerize PET. The exploration of several experimental conditions never investigated so far revealed that TPA and DMT were produced in this process, besides the expected DET and EG. The different primary and secondary reactions involved allowed directing the process to the production of different monomers by changing the reaction conditions. TPA formation was promoted at short time subcritical ethanolysis treating high % wt. DET was produced as a primary product of depolymerization and furthermore as a secondary product from the side reaction between TPA and ethanol; hence, its production was triggered under long supercritical ethanolysis.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acssuschemeng.1c02489>.

Extended information about analytical procedures and calculation of error bars; spectrophotometric analyses after NaOH extraction of different solids; SEM complete characterization of commercial and postreaction pellets; and reaction of compressed ethanol with standard TPA, DMT, and EG (PDF)

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Notes

The authors declare no competing financial interest.

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